

Marian Temprana Alonso

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Professional Summary

AI software engineer and researcher with a PhD in Computer Science, building private, enterprise-grade generative AI systems in an R&D group. Hands-on experience with LLM fine-tuning, retrieval-augmented generation, multimodal guardrails, and image and video generation pipelines, deployed on Kubernetes and Azure. Background in machine learning, optimization, and information theory, with a track record of turning research ideas into benchmarked, working systems and presenting them to technical and client audiences.

Skills

Programming: Python (advanced), PyTorch (advanced), TensorFlow (proficient), MATLAB (proficient)

Generative AI: LLM fine-tuning (LoRA/QLoRA), RAG, multi-agent systems and tool-calling, prompt engineering, multimodal guardrails and red-teaming, image and video generation, benchmarking and LLM-as-judge evaluation

LLM & Model Tooling: LangChain, LangGraph, Hugging Face, vLLM, vLLM-Omni, Ollama, Diffusers

Infrastructure & MLOps: Azure, Kubernetes, Docker, multi-GPU serving, quantization and inference optimization, automated fine-tuning pipelines, REST APIs, Git, Linux

ML & Modeling: Deep Learning, Computer Vision (YOLO, MiDaS), Graph Neural Networks, Reinforcement Learning, Optimization, Statistical Modeling, Information Theory

Scientific Python: NumPy, SciPy, Jupyter

Languages: English (bilingual), Spanish (native)

Professional Experience

United Data Technologies

AI Software Engineer, R&D Group

Miramar, FL

Feb 2026 – Present

- Build components of a private enterprise AI platform, in production with clients and used internally at UDT, running LLM and multimodal AI on each organization's own data and infrastructure
- Developed automated pipelines for fine-tuning LLMs (LoRA/QLoRA) on company data; implemented production RAG across multiple isolated workspaces, each with large document collections, plus multi-agent, tool-calling systems built with LangChain and LangGraph
- Designed and deployed multilevel, multimodal (text, image, audio) guardrails covering 14 harm categories, with red-teaming and safety datasets to evaluate them
- Built image and video generation pipelines and benchmarking tools, including LLM-as-judge evaluation of generative model quality
- Served and optimized models from 4B to 120B parameters (quantization, throughput and latency tuning, multi-GPU) with vLLM, vLLM-Omni, Ollama, Hugging Face and Diffusers on Kubernetes and Azure, exposed through internal APIs and services
- Present technical work to clients and contribute to client working sessions

Sandia National Laboratories

Graduate Research Intern

Albuquerque, NM

May 2023 – Jul 2025

- Applied data mining and advanced analytics to post-process large-scale, high-contingency power grid simulations, in collaboration with Los Alamos National Laboratory
- Identified cascading failure patterns and system vulnerabilities with direct relevance to grid resilience and contingency planning
- Developed regression-based predictive models to assess topology changes and mitigate the impact of High-Altitude Electromagnetic Pulses (HEMPs) on the grid

Florida International University

Graduate Research Assistant

Miami, FL

Aug 2022 – Dec 2025

- Conducted applied research combining machine learning, optimization, and information-theoretic methods for sensing and communication systems, in multi-institution projects spanning perception, sensor fusion, and communication-efficient learning, with collaborators at NYCU, IIT Kanpur, Arizona State, Weizmann Institute, University of Melbourne, and UC Irvine

Applied Machine Learning, Perception & Reasoning

- Built end-to-end pipelines for traffic scene understanding combining object detection, depth estimation, and semantic reasoning
- Used large language models to generate structured representations of visual scenes for downstream reasoning tasks
- Integrated temporal knowledge graphs with graph neural networks to support prediction and analysis in dynamic environments

Modeling, Optimization & Communication Systems

- Designed reinforcement learning-based approaches to approximate optimal quantization strategies
- Implemented mutual information approximation and optimization methods for communication and sensing problems
- Evaluated models through large-scale simulations in wireless communication and task-based quantization settings

Education

PhD	2025
Florida International University, Computer Science	
<i>Dissertation: An Information-Theoretic Framework for Energy-Efficient Communication, Compression, and Inference</i>	
MSc	2025
Florida International University, Computer Science	
BSc	2022
Florida International University, Applied Mathematics	

Selected Honors & Awards

Honorary member of IEEE Information Theory Society Student and Outreach (S&O) subcommittee	2025
ISIT Travel Grant	2024
ISIT DeepVerse 6G Machine Learning Challenge Winning Team	2023
GAAN Fellowship	2023

Publications & Pre-Prints

[J2] S. Srivastava, **M. Temprana Alonso**, R. Chatterjee, N. Askar, U. Demirhan, F. Shirani, S. Rini, A. Alkhateeb. “Deep Learning for Multi-Modal Sensor Fusion and CSI Compression in Vehicular Communications.” *IEEE Transactions on Vehicular Technology*, 2025.

[J1] **M. Temprana Alonso**, X. Liu, H. Aghasi, F. Shirani. “Capacity Gains in MIMO Systems with Few-Bit ADCs Using Nonlinear Analog Circuits.” Under review for *IEEE Journal on Selected Areas in Information Theory*, 2025. <https://arxiv.org/abs/2212.05948>

[C2] **M. Temprana Alonso**, F. Shirani, N. Irwin Bernardo, Y.C. Eldar. “Non-Linear Analog Processing Gains in Task-Based Quantization.” *IEEE International Symposium on Information Theory (ISIT)*, 2024.

[C1] **M. Temprana Alonso**, F. Shirani, S.S. Iyengar. “Optimal Fault-Tolerant Data Fusion in Sensor Networks: Fundamental Limits and Efficient Algorithms.” *IEEE Information Theory Workshop (ITW)*, 2023.

[C3] S. Srivastava, **M. Temprana Alonso**, N. Askar, U. Demirhan, F. Shirani, S. Rini, A. Alkhateeb. “Deep Learning-Based Multi-Modal Sensor Fusion for CSI Compression in Wireless Networks.” *IEEE ICMLCN*, 2025.